Alexander Tsai

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EXPERIENCE

Machine Learning Engineer Intern Sep 2026 – Present
TikTok San Jose, CA

- Developed a classifier model to identify **AI-generated images** and prevent refund abuse, achieving **99.85% precision / 76% recall** on production data and 66% recall on out-of-distribution generators.
- Trained **sparse autoencoders** and applied attribution heatmaps to bolster model interpretability for reviewers.
- Deployment globally led to a projected reduction of **$4M** in refund abuse losses annually.

Quantitative Research Intern May 2026 – Aug 2026
ExodusPoint Capital Management New York, NY

- Developed a novel credit market trading signal by performing **multilinear regression analysis** on six years of intraday historical data.
- Applied hypothesis testing to determine the incremental value of an external dataset over public data, saving **$75k** in annual costs.
- Built a model to infer trade direction and volume from execution prices in public order books using Weibull curves and decision trees.

Machine Learning Engineer Intern Jul 2025 – May 2026
TikTok San Jose, CA / Singapore

- Built and deployed a **content moderation pipeline** using Qwen3-VL, vLLM, and XGBoost, enforcing **4k+ accounts** daily.
- Fine-tuned Qwen3-VL using **GRPO reinforcement learning** on a curated dataset, improving video feature extraction accuracy by 12%.
- Increased **inference throughput by 5.35x** over default configurations via batching, asynchronous inference, and multithreading.

Data Science Intern May 2025 – Jul 2025
Tradeweb New York, NY

- Built an XGBoost model to predict transaction costs using 5 years of historical trade data totalling **$4.7T** in notional volume.
- Reduced RMSE by **21%** over baseline via **hyperparameter optimisation** and SHAP-guided feature engineering.

EDUCATION

Singapore Management University Singapore
B.S. in Human-Computer Interactions Aug 2023 – May 2027 (expected)

- Dean's List AY2024-2026

University of California, Berkeley Berkeley, CA
Visiting Student under Global Computing and Data Science Program Aug 2026 – Dec 2026

RESEARCH

Research Fellow May 2024 – May 2025
Reach Alliance Singapore

- Developed a multilingual **RAG** chatbot using **LangChain** and Firestore to broaden financial literacy access for migrant workers; findings published and presented at an annual University of Toronto conference.
- Presented research as a **TEDx** speaker: *Behavioural Sciences and Conversational AI*.

LEADERSHIP

AI For Youth Instructor Aug 2024 – Dec 2024
Sustainable Learning Lab 2 Singapore

- Instructed 30 students under **Intel's AI for Youth Program**, teaching Python, statistics, and neural networks.
- Guided 6 teams on capstone projects, including computer vision–based attendance systems and emotional-support chatbots.

SKILLS

- **Programming**: Python, Bash, R, SQL
- **ML Libraries**: PyTorch, vLLM, Transformers, scikit-learn, OpenCV, LangChain
- **Infrastructure & Data**: Docker, Kubernetes, Spark, Polars, Pandas, NumPy, Git
- **Interests**: Drums, Neovim, Snowboarding